

LINEAR PATH POWER POSTERIOR





e.g. if the data is iid

$$L(y \mid x)^\beta = \prod_i L(y_i \mid x)^\beta$$

► $\beta < 1$ synthetically downweights data

▶ 1 synthetically identifying outliers in data

▶ We can re-interpret the linear path in terms of Bayesian terminology:

► When $L_{\beta}(y | x) = L(y | x)^{\beta}$ we obtain the power posterior:

$$p_{\beta}(x | y) = \frac{L(x | y)^{\beta} p(x)}{p_{\beta}(y)}$$



Has applications in robust Bayesian statistics, and crests, etc

$$= \frac{w^{\beta}(x)\eta(x)}{Z(\beta)} = \pi(x)$$

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$$p_\beta(x | y) = \frac{L(x | y)^\beta p(x)}{p_\beta(y)} = \frac{w^\beta(x) \eta(x)}{Z(\beta)} = \pi(x)$$

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- ▶ e.g. if the data is iid

$$L(y | x)^\beta = \prod_i L(y_i | x)^\beta$$

- ▶ $\beta < 1$ synthetically downweights data
- ▶ $\beta > 1$ synthetically identifying outliers in data
- ▶ Has applications in robust bayesian statistics, and coresets, etc

ANNEALING AS GLOBAL OPTIMISATION